

Muhammad Adeel

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EDUCATION

Bachelor in Computer Science, University of Haripur

Sep 2021 - June 2025

- Major in Computer Science.
- Final Year Project: "Spine Disease Detection System".

Faculty of Computer Science, FG Science Degree College Wah Cantt

2019 - 2021

PROFESSIONAL EXPERIENCE

Data Analyst | Positive Solutions, Wah Cantt

April 2026 - July 2026

- Analyzed and processed large datasets using Excel, MySQL, and Python to support data-driven business decisions across departments.
- Built and maintained interactive reports and dashboards to track key performance metrics and monitor business trends over time.
- Wrote and optimized SQL queries to extract, clean, and organize data from company databases, improving reporting accuracy and turnaround time.
- Automated repetitive data-processing tasks using Python and leveraged AI tools to streamline analysis and support faster decision-making.

Python Development Trainee | NAVTTC (Training), Islamabad

Oct 2025 - March 2026

- Assisted in the development and testing of basic machine learning models, contributing to data preparation and evaluation stages.
- Worked extensively with Python libraries such as NumPy and Pandas for data manipulation, cleaning, and analysis.
- Participated in data preprocessing and visualization tasks to support model training and performance assessment.
- Gained hands-on exposure to end-to-end AI/ML workflows while collaborating with a team on real project tasks.

PROJECTS

Spine Disease Detection System | CNN (EfficientNetV2S)

- Developed a computer vision-based medical imaging system to detect and classify spinal disorders (Neural Foraminal Narrowing, Subarticular Stenosis, Spinal Canal Stenosis) using the RSNA Lumbar Spine dataset.
- Fine-tuned an EfficientNetV2S architecture achieving 95% classification accuracy across disorder categories, with image preprocessing and augmentation to improve generalization on DICOM scans.

Third Eye — AI-Powered Assistant for Visually Impaired Persons

- Built a real-time multi-module assistive system combining scene description (BLIP), object detection (YOLO11, 80 COCO classes), text extraction (EasyOCR), and Pakistani currency recognition (fine-tuned DenseNet121).
- Integrated text-to-speech (gTTS) to convert the combined analysis into real-time spoken feedback, helping visually impaired users understand surroundings, read text, and identify currency notes.

Heart Disease Prediction | Ensemble ML Web App

- Built a web app predicting heart disease likelihood on the Kaggle Playground Series S6E2 dataset (based on UCI Cleveland Heart Disease data) via EDA, correlation analysis, and an ensemble of XGBoost, LightGBM, and CatBoost models.
- Achieved an ensemble AUC of ~0.956 on binary classification of disease presence/absence using averaged model probabilities.

Movie Recommender System | Streamlit

- Developed a content-based movie recommendation web app using Streamlit, suggesting similar movies based on genre, cast, and plot features for a user-searched title.

Multimodal Parkinson's Disease Detection Using Voice (Research Project - In Progress)

- Designing a multimodal cross-attention framework combining patient audio and speech transcripts, using a LoRA fine-tuned Wav2Vec2 model to detect Parkinson's disease while reducing memory overhead and catastrophic forgetting.
- Applying GroupKFold cross-validation to prevent data leakage and SHAP-based interpretability to explain model predictions for clinical trust.

HONOURS AND AWARDS

Top 5th Best Final Year Project - Department of Information Technology, University of Haripur

CERTIFICATIONS

- Machine Learning Specialization - Andrew Ng (Coursera)
- Deep Learning Specialization - Andrew Ng (Coursera)
- Python Data Structures - University of Michigan (Coursera)

CONFERENCES & SEMINARS

- Machine Learning Bootcamp - University of Haripur | The Robotics with AI Seminar